

Individual Pathways

Activity 6.1

Revising the atom

1. Identify the set that contains an element, a compound and a mixture.
 - (a) carbon dioxide, air, silver
 - (b) gold, sulfur, phosphorus
 - (c) granite, oxygen, hydrogen
 - (d) copper oxide, calcium carbonate, sodium chloride
2. Identify the correct statement.
 - (a) Niels Bohr demonstrated that atoms combine in whole number ratios.
 - (b) Democritus discovered the atomic nucleus.
 - (c) J.J. Thomson discovered the electron.
 - (d) Rutherford discovered the neutron.
3. Select the correct statement about the atom.
 - (a) Protons have the same electrical charge as neutrons.
 - (b) Rutherford used alpha particles from a radioactive element to investigate the sub-structure of the atom.
 - (c) The hydrogen-1 isotope consists of one proton, one neutron and one electron.
 - (d) All the electrons in an atom are located at the edge of the spherical atom.
4. Select the correct statement about hydrogen isotopes.
 - (a) H-2 contains two protons, two neutrons and one electron
 - (b) H-1 contains one neutron and one electron
 - (c) H-1 contains one proton and one neutron
 - (d) H-3 contains one proton, two neutrons and one electron
5. Select the statement that is true of carbon-14.
 - (a) Its nucleus contains six neutrons.
 - (b) It has one more neutron than carbon-12.
 - (c) It has two more electrons than carbon-12.
 - (d) It is radioactive.

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6. Beta particles are
 - (a) fast-moving electrons.
 - (b) positively charged.
 - (c) heavier than an alpha particle.
 - (d) unable to penetrate human skin.

7. The radioactive element present in smoke detectors is
 - (a) americium-241.
 - (b) uranium-238.
 - (c) strontium-90.
 - (d) carbon-14.

8. Gamma rays consist of
 - (a) streams of neutrons.
 - (b) high energy electromagnetic radiation.
 - (c) positive protons.
 - (d) negative particles travelling at 85% of the speed of light.

9. In nuclear symbolism, the letter 'Z' stands for
 - (a) the atomic number — the number of neutrons in the nucleus.
 - (b) the atomic number — the number of protons in the nucleus.
 - (c) the number of neutrons.
 - (d) the mass number.

10. Consider the nuclear symbol ${}_{92}^{235}\text{U}$. Select the true statement relating to this symbol.
 - (a) Uranium atoms contain more protons than neutrons.
 - (b) There are 235 electrons orbiting the nucleus.
 - (c) The atomic number of uranium is 235.
 - (d) The nucleus contains 143 neutrons.

11. Cosmic radiation mainly consists of
 - (a) X-rays.
 - (b) infra-red rays.
 - (c) high energy protons.
 - (d) electrons.

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12. Iodine -131 has a half-life of 8 days. If a sample of I-131 has a mass of 60 g, how much radioactive iodine will be left after 24 days?
- (a) 7.5 g
 - (b) 15 g
 - (c) 30 g
 - (d) 3.75 g
13. The symbol for a gamma ray is
- (a) α
 - (b) λ
 - (c) β
 - (d) γ
14. Answer true or false for each statement.
- (a) Ammonia is a compound.
 - (b) Calcium is an element.
 - (c) John Dalton proposed that matter contained positive and negative charges.
 - (d) Isotopes have different mass numbers.
 - (e) H-3 is a radioisotope.
15. Answer true or false for each statement.
- (a) Carbon-12 has six neutrons.
 - (b) Alpha particles contain two protons and two neutrons.
 - (c) Rutherford's atomic model had electrons orbiting the nucleus.
 - (d) Bohr's atomic model had electrons and neutrons orbiting the nucleus.
 - (e) Neutrons have a negative charge.
16. Answer true or false for each statement.
- (a) Radioactive elements are often stored in lead boxes.
 - (b) Alpha particles pose a great external danger to the body.
 - (c) There are about 50 natural radioisotopes.
 - (d) Beta particles travel about 100 cm through the air.
 - (e) Becquerel accidentally discovered radioactivity while investigating uranium compounds.